

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected, May 2028

East Lansing, MI

- **GPA:** 3.5 / 4.0
- **Relevant Coursework:** Linear Algebra, Discrete Mathematics, Object-Oriented Software Design, Computer Organization and Architecture, Data Structures & Algorithms

EXPERIENCE

Claude Builder Club @ MSU

Vice President & Co-Founder

Oct. 2025 – Present

East Lansing, MI

- Eliminated manual onboarding entirely by automating member provisioning (GitHub API + Slack SDK) to grant full repo and workspace access in 30 seconds, cutting setup time 97% across 100+ members.
- Architected event-driven attendance infrastructure using QR token generation backed by AWS S3 caching, replacing manual sign-in with real-time tracking and cutting processing overhead by 80%.
- Engineered GitHub Actions CI/CD pipeline with enforced 90% coverage gates across 30+ tests and branch protection rules on all production merges, reducing bug escape rate by 65%.

MindMosaic

Software Engineer

Aug. 2025 – Nov. 2025

East Lansing, MI

- Designed polyglot persistence architecture (PostgreSQL + Neo4j) with environment-parity Docker Compose configuration, enabling one-command local setup and eliminating configuration drift between dev and staging.
- Eliminated graph query bottlenecks by refactoring Cypher queries with relationship-type indexing, cutting p95 latency 48% (850ms → 440ms) and making recursive traversals 3× faster than equivalent SQL joins.
- Maintained 85% test coverage across JUnit and Jasmine suites via GitHub Actions CI/CD with branch protection, preventing untested code from reaching production on all merges.

PROJECTS

Mira — Agentic Voice Assistant | *FastAPI, AWS Lambda, Redis, Celery, Docker* May 2025 – Nov. 2025

- Deployed containerized FastAPI inference service on AWS Lambda with lazy-loaded ML models (Whisper, Sentence-BERT, spaCy), eliminating idle compute costs and reducing cold-start latency 86% (6,643ms → 930ms).
- Decoupled ML inference from the API layer by offloading the RAG pipeline to an async Redis/Celery task queue, cutting end-to-end response time by 73% and enabling horizontal worker scaling under load.
- Built bidirectional IPC event bus between Electron and Next.js client layers with a dedicated webhook server, achieving sub-50ms event propagation and speaker transcription delivery under 2 seconds.

WizViz — SpartaHack X | *Python, MediaPipe, OpenCV, NumPy, TensorFlow*

Feb. 2025

- Won Interactive Media Track: built a gesture-controlled wizard duel where real-time full-body pose estimation drives spell casting, blocking, and AI combat — entirely in a browser with no game engine.
- Built full-body gesture pipeline using MediaPipe Pose to map 33 skeletal landmarks into game actions; optimized inference loop to sustain 60 FPS with sub-20ms input lag.
- Developed custom 2D vector physics engine with AABB collision detection to translate live 3D skeletal coordinates into game state, maintaining <5px drift at 60 FPS.

TECHNICAL SKILLS

Languages Python, Java, JavaScript/TypeScript, SQL, Cypher, Golang, C++, Bash
Frameworks FastAPI, Spring Boot, Node.js, React, Next.js, Celery
Databases PostgreSQL, Neo4j, MongoDB, Redis
Cloud & Infrastructure AWS (Lambda, EC2, RDS, S3), Docker, GitHub Actions, CI/CD Pipelines